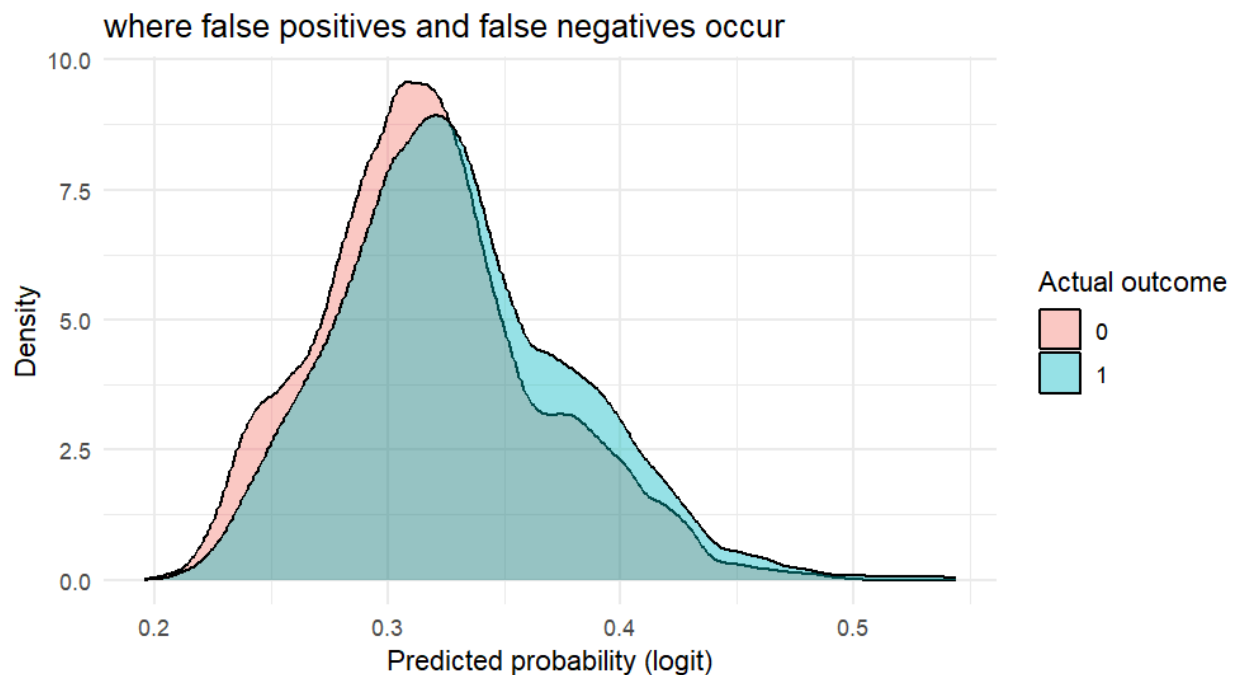
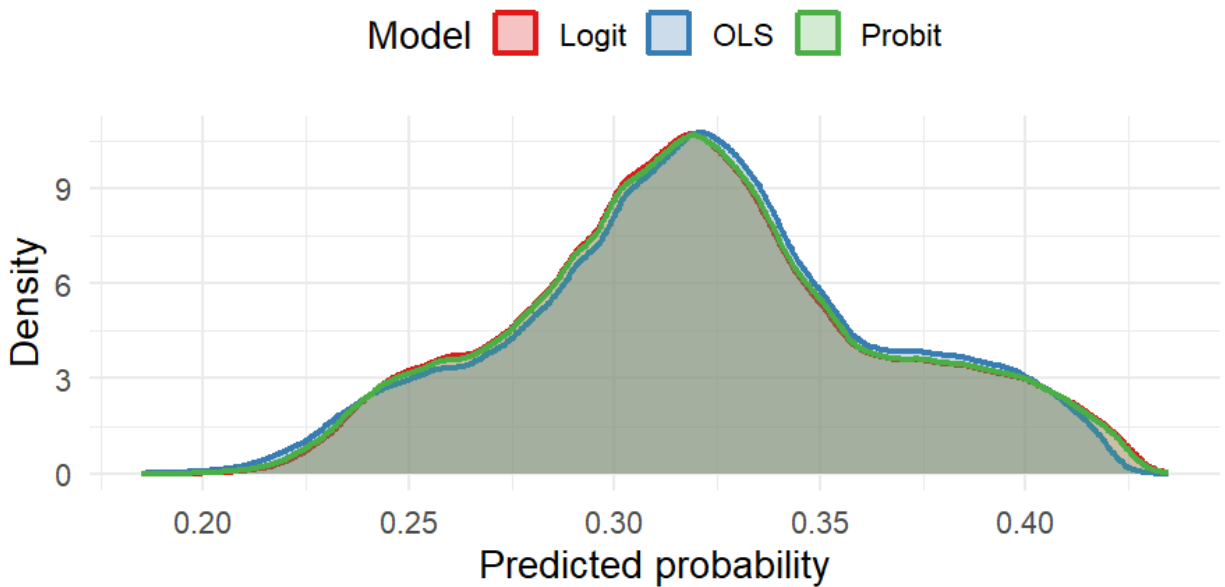


Distribution of Predicted Probabilities Across Models

OLS vs Logit vs Probit



The distributions overlap a lot making it harder to predict. Many couples who are age-gap (1) have predicted probabilities similar to couples who are not (0). The model cannot separate them. The distribution of predicted probabilities across the models produce very similar predicted probability distributions. This shows the marginal effect of age + education is small.

```
library(ggplot2)
library(tidyverse)
library(haven)
library(tidyr)
```

```
load("ACS_2021_couples.RData")
```

```
acs2021_couples$RACE <- fct_recode(as.factor(acs2021_couples$RACE),
  "White" = "1",
  "Black" = "2",
  "American Indian or Alaska Native" = "3",
  "Chinese" = "4",
  "Japanese" = "5",
  "Other Asian or Pacific Islander" = "6",
  "Other race" = "7",
  "two races" = "8",
  "three races" = "9")
```

```
acs2021_couples$h_race <- fct_recode(as.factor(acs2021_couples$h_race),
  "White" = "1",
  "Black" = "2",
  "American Indian or Alaska Native" = "3",
  "Chinese" = "4",
  "Japanese" = "5",
  "Other Asian or Pacific Islander" = "6",
  "Other race" = "7",
  "two races" = "8",
  "three races" = "9")
```

```
acs2021_couples$HISPAN <- fct_recode(as.factor(acs2021_couples$HISPAN),
  "Not Hispanic" = "0",
  "Mexican" = "1",
  "Puerto Rican" = "2",
  "Cuban" = "3",
  "Other" = "4")
```

```
acs2021_couples$h_hispan <- fct_recode(as.factor(acs2021_couples$h_hispan),
  "Not Hispanic" = "0",
  "Mexican" = "1",
  "Puerto Rican" = "2",
  "Cuban" = "3",
  "Other" = "4")
```

```
trad_data <- acs2021_couples %>% filter( (SEX == "Female") & (h_sex == "Male") )
```

```
trad_data$he_more_than_3yrs_than_her <- as.numeric(trad_data$age_diff < -3)
```

```
table(trad_data$he_more_than_3yrs_than_her, cut(trad_data$age_diff, c(-100, -10, -3, 0, 3, 10, 100)))
```

```
trad_data <- trad_data %>%
  mutate(educ_numeric = case_when(
```

```
educ_advdeg == 1 ~ 18, educ_college == 1 ~ 16, educ_somcoll == 1 ~ 14, educ_hs == 1 ~ 12, educ_nohs == 1 ~ 10,
  TRUE ~ NA_real_))
```

```
trad_data <- trad_data %>%
  mutate(
    h_educ_chr = as.character(h_educ),
    h_educ_numeric = case_when(
      grepl("Graduate|Master|Professional|Doctor", h_educ_chr, ignore.case = TRUE) ~ 18,
      grepl("Bachelor", h_educ_chr, ignore.case = TRUE) ~ 16,
      grepl("Some college|Associate", h_educ_chr, ignore.case = TRUE) ~ 14,
      grepl("High school", h_educ_chr, ignore.case = TRUE) ~ 12,
      grepl("Less than|No schooling|Grade", h_educ_chr, ignore.case = TRUE) ~ 10,
      TRUE ~ NA_real_))
```

```
ols_out1 <- lm(he_more_than_3yrs_than_her ~ INCWAGE + educ_college + educ_advdeg + AGE, data = trad_data)
```

```
pred_vals_ols1 <- predict(ols_out1, trad_data)
pred_model_ols1 <- (pred_vals_ols1 > mean(pred_vals_ols1))
table(pred = pred_model_ols1, true = trad_data$he_more_than_3yrs_than_her)
```

```
model_logit1 <- glm(he_more_than_3yrs_than_her ~ educ_hs + educ_somcoll + educ_college + educ_advdeg + AGE, data = trad_data, family = binomial)
summary(model_logit1)
```

```
pred_vals <- predict(model_logit1, trad_data, type = "response")
pred_model_logit1 <- (pred_vals > mean(pred_vals))
table(pred = pred_model_logit1, true = trad_data$he_more_than_3yrs_than_her)
```

```
ols_out1 <- lm(he_more_than_3yrs_than_her ~ INCWAGE + educ_college + educ_advdeg + AGE,
  data = trad_data)
```

```
logit_out1 <- glm(he_more_than_3yrs_than_her ~ INCWAGE + educ_college + educ_advdeg + AGE,
  data = trad_data, family = binomial(link = "logit"))
```

```
probit_out1 <- glm(he_more_than_3yrs_than_her ~ INCWAGE + educ_college + educ_advdeg + AGE,
  data = trad_data, family = binomial(link = "probit"))
```

```
pred_df <- tibble(
  OLS = predict(ols_out1, trad_data),
  Logit = predict(logit_out1, trad_data, type = "response"),
  Probit = predict(probit_out1, trad_data, type = "response")) %>%
  pivot_longer(cols = everything(),
    names_to = "Model",
    values_to = "Predicted")
```

```
ggplot(pred_df, aes(x = Predicted, fill = Model, color = Model)) +
  geom_density(alpha = 0.25, linewidth = 1) +
  scale_fill_brewer(palette = "Set1") +
  scale_color_brewer(palette = "Set1") +
  labs(
    title = "Distribution of Predicted Probabilities Across Models",
    subtitle = "OLS vs Logit vs Probit",
    x = "Predicted probability",
    y = "Density"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "top")
```

